FROM USER INTENT TO THE SYSTEM MODEL

The missing link between UX and systems engineering

48 hours to build a universal UX/Engineering translator
connecting UserShaker, SysML v2 and CATIA Magic

HACKATHON · 2026

ESILV Capgemini DASSAULT SYSTEMES Renault Group





THE NIGHTMARE SCENARIO

1 A simple intention

“I want to plan my trip while charging. Simple. Seamless. Right the first time.”

2 The digital game of telephone

UX, requirements, workshops, documents, models: every handoff adds noise.

3 The silent crash

In the end, no one knows whether the system still reflects the original need.

Your mission: cut the noise before it turns into cost, delay… or a bad car.

THE PROBLEM: HIGH-RISK TRANSLATION

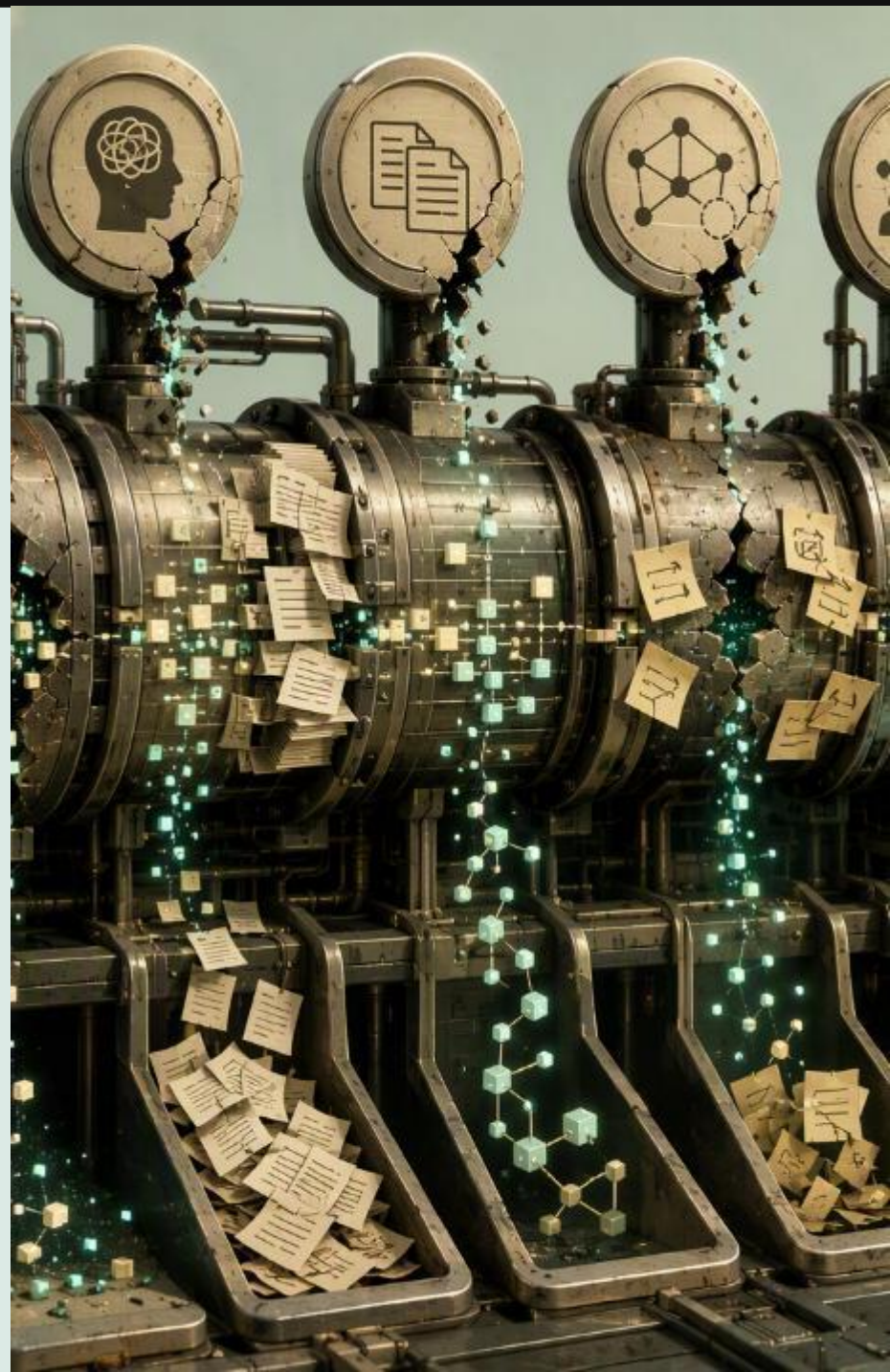
Fragmented intent — UX data is not structured for engineering.

Interpretation bias — Every rewording can distort the original intent.

Blind spots — Actors, constraints and implicit requirements can disappear.

Broken traceability — No quick way to prove which need is covered.

A fast translation that deletes one word in ten without warning you. Still confident?



WHAT YOU WILL GAIN IN 48 HOURS

A REAL PROBLEM. FOUR FRONTIERS. ONE DEMO TO DEFEND.

01 THINK PRODUCT

Decide what must carry over from user intent to the system model.

02 DESIGN THE ARCHITECTURE

Design a simple, robust and explainable pipeline.

03 MASTER DEEP TECH

Combine AI, semantics, MBSE, SysML v2 and traceability.

04 PITCH LIKE A STARTUP

Convince Renault with evidences, not promises.

You will not just code a prototype. You will turn industrial ambiguity into a value proposition.

WHY YOU WILL LOVE IT!

Real deep tech

The buzzwords must work: AI, MBSE, SysML v2, parsing, traceability.

A problem with no answer key

Choose, test, fail fast and defend a credible architecture.

A real industrial customer

Act like a startup pitching its concept to Renault.

In 48 hours, you are not doing an assignment. You are building a demo that can start an industrial conversation.



HACKATHON MISSION: 48 HOURS TO CHANGE THE GAME

01

INGEST

Read UserShaker data and preserve actors, journeys, context and intent.

02

TRANSLATE

Generate a consistent, structured and traceable SysML v2 model.

03

PROVE

Import and visualize the result in CATIA Magic.

BONUS LEVEL

“Chat with your model”: query the model with an LLM to detect uncovered needs, inconsistencies and omissions.

Deliverable: a working demonstrator, not just a promise.





CALL TO ACTION

Think startup

A clear vision. A simple architecture. Compelling proof.

Aim for the “wow”

Show the end-to-end transformation in just a few minutes.

Earn trust

Show where information is preserved, verified and traced.

WHO WANTS TO BUILD THE ChatGPT OF SYSTEMS ENGINEERING?

Build your team. Take the data. Make the break disappear.



IN 48 HOURS...

A user intent can become an actionable system model.

“Industry automated factories.
AI is automating code.
Now it is your turn to automate the journey from idea to system.”

The next breakthrough starts here.

Confidential C

Prerequisites

01 Specifications / documentation

- 2x SysML V2 cheatsheets + API Spec: https://www.omg.org/sysml/sysmlv2/
- UserShaker : DataModel specification document

02 Tools

- Download and install **CATIA Magic SysML v2 Community Edition**
- https://discover.3ds.com/free-catia-sysmlv2-community-edition

03 Dataset

- Usershaker : To be provided by UserShaker + explanatory deck when the hackathon starts
- SysML V2 API (Python) : https://github.com/Systems-Modeling/SysML-v2-API-Python-Client

04 Coding

- VibeCoding, Gen AI (Python, Rust…)… proceed as to wish !
- Ask AI !

On-site support by UserShaker specialist on Nov. 4th

Available for you, don’t miss the opportunity !